
PHYSICS

9702/33

Paper 3 Advanced Practical Skills 1

October/November 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **7** printed pages.



Cambridge Assessment
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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Question	Answer	Marks
1(a)	Value of I with unit and in the range $0.010\text{ A} \leq I \leq 0.200\text{ A}$.	1
1(b)	Value of y with unit and in the range $20.0\text{--}75.0\text{ cm}$.	1
1(c)	Six sets of readings of P , Q (different pairs) and y without help from the Supervisor scores 4 marks, five sets scores 3 marks etc.	4
	Range: Resistor values include the pair (15Ω and 12Ω) and the pair (27Ω and 22Ω).	1
	Column headings: Each column heading must contain a quantity, a unit and a separating mark where appropriate. The presentation of quantity and unit must conform to accepted scientific convention e.g. y/m and $PQ/(P+Q)/\Omega$.	1
	Consistency: All values of y must be given to the nearest mm only.	1
	Significant figures: Values of $PQ/(P+Q)$ must be given to 2 or 3 significant figures.	1
	Calculation: Values of $PQ/(P+Q)$ are correct.	1
1(d)(i)	Axes: Sensible scales must be used, no awkward scales (e.g. $3:10$ or fractions). Scales must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions. Scales must be labelled with the quantity that is being plotted. Scale markings should be no more than three large squares apart.	1
	Plotting of points: All observations in the table must be plotted on the grid. Diameter of plotted points must be $\leq$ half a small square (no “blobs”). Points must be plotted to an accuracy of half a small square.	1
	Quality: General trend of points on graph must be negative. All points in the table (at least 5) must be plotted on the grid. It must be possible to draw a straight line that is within $\pm 0.050\text{ m}$ (to scale) on the y -axis of all plotted points.	1

Question	Answer	Marks
1(d)(ii)	Line of best fit: Judge by balance of all points on the grid about the candidate's line (at least 5 points). There must be an even distribution of points either side of the line along the full length. Allow one anomalous point only if clearly indicated by the candidate (i.e. circled or labelled). Line must not be kinked or thicker than half a small square.	1
1(d)(iii)	Gradient: The hypotenuse of the triangle used must be greater than half the length of the drawn line. The method of calculation must be correct. Do not allow $\Delta x / \Delta y$. Both read-offs must be accurate to half a small square in both the x and y directions. Sign of gradient on answer line must match graph.	1
	y-intercept: Correct read-off from a point on the line substituted into $y = mx + c$. Read-off must be accurate to half a small square in both x and y directions. or Intercept read directly from the graph with read-off at $x = 0$, accurate to half a small square.	1
1(e)	Value of $M = -$ candidate's gradient and value of $N =$ candidate's intercept. The values must not be fractions.	1
	Units for M (e.g. $\text{m}\Omega^{-1}$ or $\text{cm}\Omega^{-1}$ or $\text{mm}\Omega^{-1}$) and N (e.g. m or cm or mm) correct.	1
1(f)	Correct calculation of E with unit.	1

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Question	Answer	Marks
2(a)	Value of S with unit in the range $0.5\text{ cm} \leq S \leq 2.5\text{ cm}$ and to the nearest mm.	1
2(b)(i)	Values of S_1 and S_2 and $S_1 > S_2$.	1
	Value of raw θ to the nearest degree and value of $\theta < 45^\circ$ on answer line.	1
2(b)(ii)	Percentage uncertainty in θ based on an absolute uncertainty of 3–10°. If repeated readings have been taken, then the uncertainty can be half the range (but not zero) if the working is clearly shown. Correct method of calculation to obtain percentage uncertainty.	1
2(b)(iii)	Correct calculation of $e_1 \cos \theta$ and $e_2 \sin \theta$.	1
2(b)(iv)	Justification for significant figures in $e_1 \cos \theta$ linked to s.f. in: ($S_1 - S$) and θ or e_1 and θ or S , S_1 and θ .	1
2(c)	Second values of S_1 and S_2 .	1
	Quality: Second values of S_1 and S_2 greater than first values of S_1 and S_2 .	1
	Second value of θ .	1
2(d)(i)	Two values of β calculated correctly.	1
2(d)(ii)	Valid comment consistent with the calculated values of β , testing against a criterion stated by the candidate.	1
2(e)	Correct calculation of k using second value of β with a correct unit.	1

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Question	Answer	Marks
2(f)(i)	A Two readings are not enough to draw a (valid) conclusion (not “not enough for accurate results”, “few readings”). B Difficult to judge the vertical. C Difficult to determine θ/angle because of parallax error or cannot see the lines defining the angle. D Difficult to determine θ/angle because cannot hold the protractor steady. E Difficult to measure S_1 or S_2 with reason e.g. springs move or mass moves. F The stands tilt/topple/slide or the bosses twist on loading. <i>1 mark for each point up to a maximum of 4.</i>	4
2(f)(ii)	A Take many readings (for different values of m) <u>and</u> plot a graph or take more values of β <u>and</u> compare (not “repeat readings” on its own). B Method to get vertical reference e.g. plumb line/use of set square on bench with metre rule/use spirit level <u>vertically</u>. C Use a larger protractor/have a protractor drawn on card at the back/use projection ideas/take a photograph and measure angle. D Clamp protractor (valid method). E Clamp ruler/take photograph with scale or ruler in the frame. F Use G-clamps/clamp the stands/use heavy weights/lower the bosses. <i>1 mark for each point up to a maximum of 4.</i>	4